

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital VLSI Testing (01CT0629)	Aim: To design and test a scan chain in a sequential circuit.	
Experiment No: 11	Date:03-04-2026	Enrolment No: 92301733056

AIM: To design and test a scan chain in a sequential circuit.

IDE: EDA Playground

THEORY:

As digital circuits grow in complexity, testing becomes more challenging, especially for sequential elements like flip-flops that store internal states. Scan chain testing is a Design-for-Testability (DFT) technique that connects flip-flops in series to form a shift register. This allows test data to be shifted into and out of internal registers, enabling fault detection using Automatic Test Pattern Generation (ATPG).

A scan chain is a DFT (Design for Testability) technique where flip-flops are connected in a chain to:

- Shift in test data (scan-in)
- Capture circuit response
- Shift out results (scan-out)

Controlled using:

- scan_en (Scan Enable)
- scan_in
- scan_out

A. VERILOG CODE:

```
// Code your design here
module scan_chain(
    input clk,
    input rst,
    input scan_en,
    input scan_in,
    input d_in,
    output scan_out,

    output [3:0] q
);

    reg [3:0] ff;
    always @(posedge clk or posedge rst)
        begin
            if(rst)
                ff<= 4'b0000;
            else begin
                if(scan_en)
                    ff <= {ff[2:0], scan_in};
```

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```

        else
            ff <= {ff[2:0], d_in};
        end
    end
end

```

```

assign q = ff;
assign scan_out = ff[3];
endmodule

```

B. Testbench

```

// Code your testbench here
// or browse Examples
`timescale 1ns/1ps
module tb_scan_chain;
    reg clk,rst;
    reg scan_en,scan_in,d_in;
    wire scan_out;
    wire [3:0] q;

    scan_chain uut(
        .clk(clk),
        .rst(rst),
        .scan_en(scan_en),
        .scan_in(scan_in),
        .d_in(d_in),
        .scan_out(scan_out),
        .q(q)
    );

    always #5 clk = ~clk;
    initial
        begin
            $dumpfile("dump.vcd");
            $dumpvars(0,tb_scan_chain);

            clk = 0;
            rst = 1;
            scan_en = 0;
            scan_in = 0;

            d_in = 0;
            #10 rst = 0;

            scan_en=0;
            d_in=1;
            #10;
            d_in=0;
            #10;

```

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```

d_in=1;
#10;

scan_en=1;
#10;
scan_en=0;
#10;
scan_en=1;
#10;
scan_en=1;
#10;

scan_en=0;
#40;

$finish;
end
endmodule

```

SIMULATION RESULTS:



CONCLUSION:

In this experiment, The scan chain for the sequential circuit was successfully designed and tested. The circuit correctly shifts data in test mode and works normally in normal mode. This experiment shows how scan chains help in easy testing and fault detection in digital circuits.

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Post Lab Exercise:

Q1. What is scan enable?

Scan enable (scan_en) is a control signal used in scan chain circuits to switch between normal operation mode and test mode.

- When scan_en = 1 → Circuit operates in scan mode, and data is shifted through flip-flops using scan_in.
- When scan_en = 0 → Circuit operates in normal mode, and data is taken from functional input (d_in).

Q2. What is scan_out?

scan_out is the output of the last flip-flop in the scan chain. It is used to observe and analyze the internal state of the sequential circuit during testing.

It helps in verifying whether the circuit is functioning correctly by shifting out stored data.

Q3. Why scan chain is used?

Scan chain is used to improve testability of sequential circuits.

- It allows easy access to internal flip-flops.
- Enables fault detection using test patterns (ATPG).
- Simplifies testing by converting sequential logic into shift-register-like behavior.
- Helps in detecting faults such as stuck-at faults and timing issues.